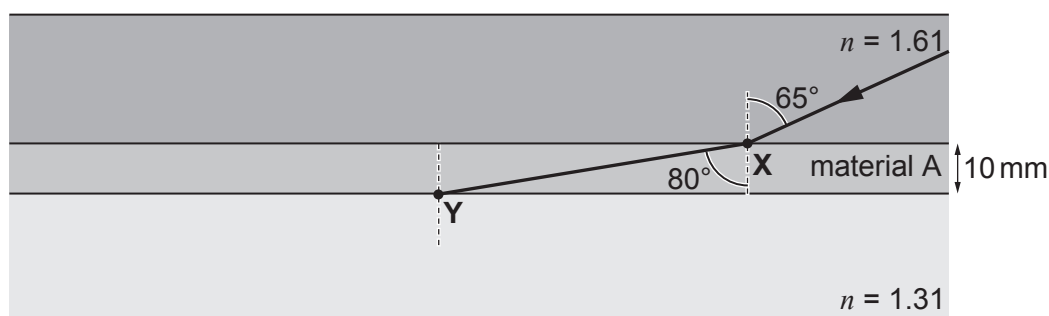


5. (a) Define the refractive index of a material. [1]

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- (b) Material A of unknown refractive index, n , is sandwiched between two rectangular blocks of different refractive indices. A ray of light travels from the top block into material A as shown.



- (i) Show that the refractive index of material A is approximately 1.5. [2]

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- (ii) Determine the time taken for the ray of light to travel through material A from X to Y. [3]

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- (c) (i) Confirm that total internal reflection will occur at Y.

[3]

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- (ii) **On the diagram opposite** continue the path of the light ray inside material A. [1]

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